Predicting Body Weight Based on Lifestyle and Demographic Factors

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1. INTRODUCTION

1.1 Background

Obesity has become a global health epidemic, with obesity rates nearly tripling worldwide since 1975 according to the World Obesity Federation. Understanding the factors that contribute to body weight is crucial for developing effective public health interventions. This study analyzes data collected from 2,111 individuals across Mexico, Peru, and Colombia to model and predict body weight based on various lifestyle, demographic, and behavioral factors.

Our dataset, sourced from the UCI Machine Learning Repository, contains comprehensive information about individuals' physical characteristics, dietary habits, and activity levels. Each observation represents a single person and includes 17 variables capturing various aspects of their lifestyle. Our primary goal is to develop a linear regression model that can accurately predict body weight using both quantitative predictors (such as age and height) and categorical predictors (such as gender and family history).

Our analysis will explore how factors including physical activity frequency, water consumption, eating habits, and transportation methods relate to body weight. By identifying significant predictors and their relationships with weight, this study aims to provide insights that could inform both individual health decisions and broader public health strategies.

1.2 Dataset Description

- **Source**: UCI Machine Learning Repository - Estimation of Obesity Levels Based on Eating Habits and Physical Condition
- **Observations**: 2,111 individuals from Mexico, Peru, and Colombia
- **Response Variable**: Weight (kg) - continuous quantitative variable
- **Predictors**: Age, Height, Gender, family history, eating habits (FAVC, FCVC, NCP), smoking (SMOKE), water consumption (CH2O), calorie monitoring (SCC), physical activity (FAF), technology use (TUE), CAEC (eating between meals), CALC (alcohol), and MTRANS (transportation)

2. METHODS

2.1 Data Cleaning

```
## No missing values detected in the dataset.
```

```
## Number of duplicate rows: 24
```

```
## Removing 24 duplicate rows...
## Rows after removing duplicates: 2087
```

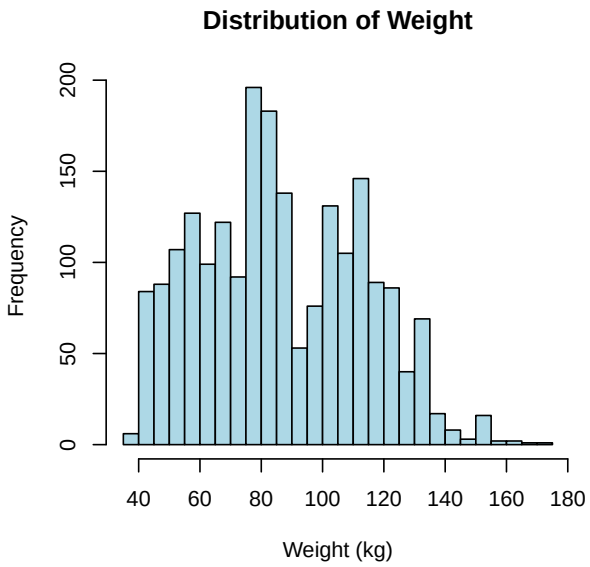
The data cleaning process revealed no missing values in our dataset of 2,087 observations. All categorical variables were properly converted to factors for appropriate handling in our regression models. All duplicate values were removed.

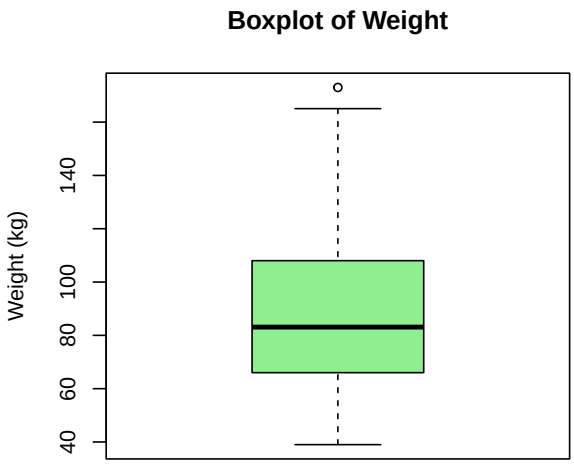
2.2 Exploratory Data Analysis (EDA)

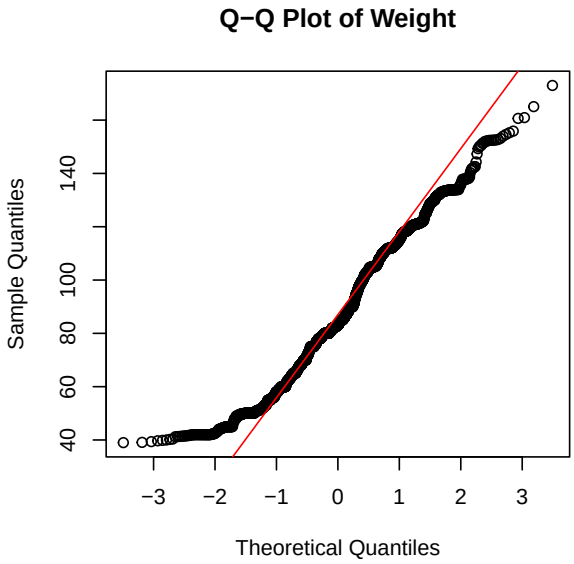
```
##       Age             Height          Weight            FCVC
##  Min.   :14.00   Min.   :1.450   Min.   : 39.00   Min.   :1.000
```

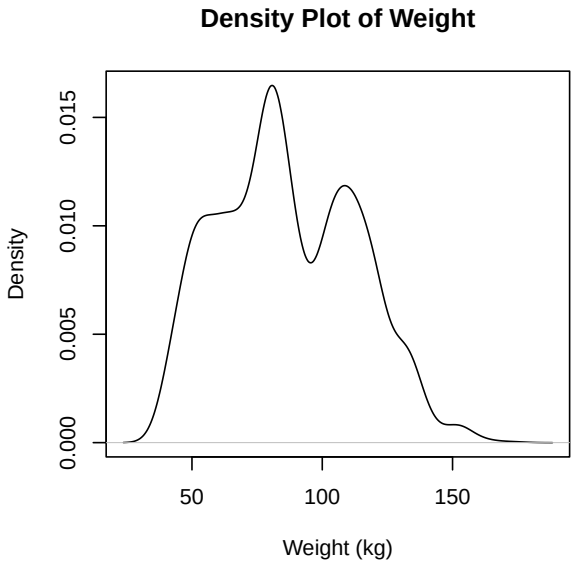
```
##  1st Qu.:19.92   1st Qu.:1.630   1st Qu.: 66.00   1st Qu.:2.000
##  Median :22.85   Median :1.702   Median : 83.10   Median :2.396
##  Mean   :24.35   Mean   :1.703   Mean   : 86.86   Mean   :2.421
##  3rd Qu.:26.00   3rd Qu.:1.769   3rd Qu.:108.02   3rd Qu.:3.000
##  Max.   :61.00   Max.   :1.980   Max.   :173.00   Max.   :3.000
##       NCP             CH2O            FAF              TUE
##  Min.   :1.000   Min.   :1.000   Min.   :0.0000   Min.   :0.0000
##  1st Qu.:2.697   1st Qu.:1.591   1st Qu.:0.1245   1st Qu.:0.0000
##  Median :3.000   Median :2.000   Median :1.0000   Median :0.6309
##  Mean   :2.701   Mean   :2.005   Mean   :1.0128   Mean   :0.6630
##  3rd Qu.:3.000   3rd Qu.:2.466   3rd Qu.:1.6781   3rd Qu.:1.0000
##  Max.   :4.000   Max.   :3.000   Max.   :3.0000   Max.   :2.0000
```

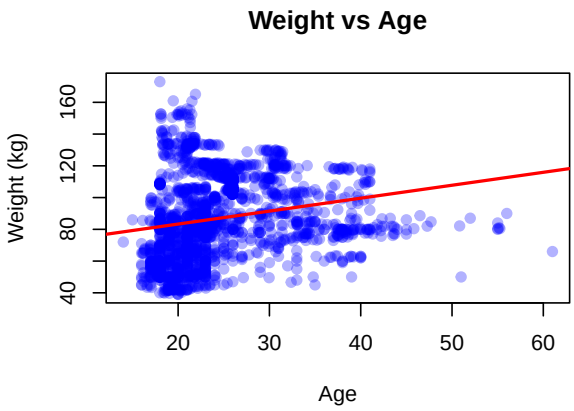
```
##
## Insufficient_Weight       Normal_Weight      Obesity_Type_I     Obesity_Type_II
##                 267                 282                 351                 297
##    Obesity_Type_III  Overweight_Level_I Overweight_Level_II
##                 324                 276                 290
```

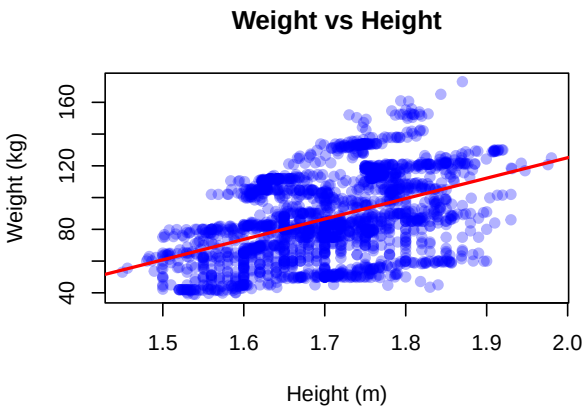

Distribution of Weight
Frequency
0
50
100
150
200
40
60
80
100
120
140
160
180
Weight (kg)

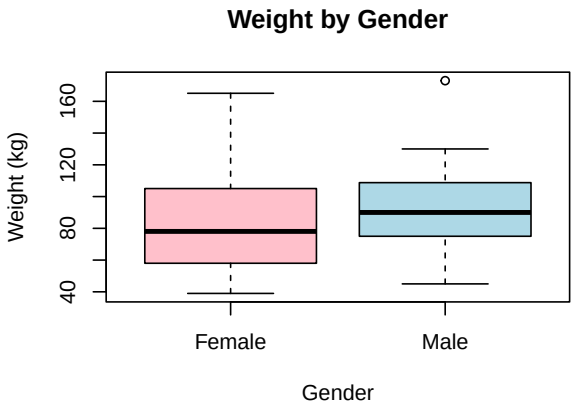

Boxplot of Weight
Weight (kg)
40
60
80
100
140

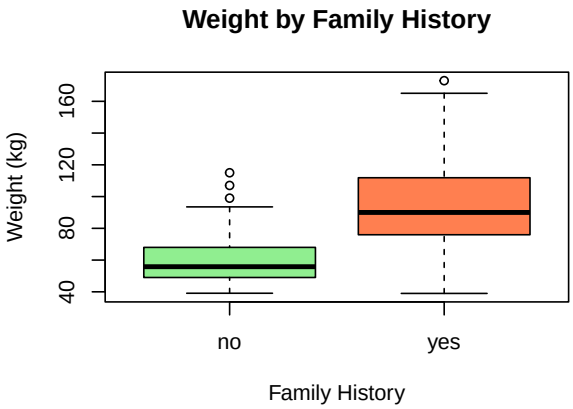

Q−Q Plot of Weight
Sample Quantiles
40
60
80
100
140
−3
−2
−1
0
1
2
3
Theoretical Quantiles

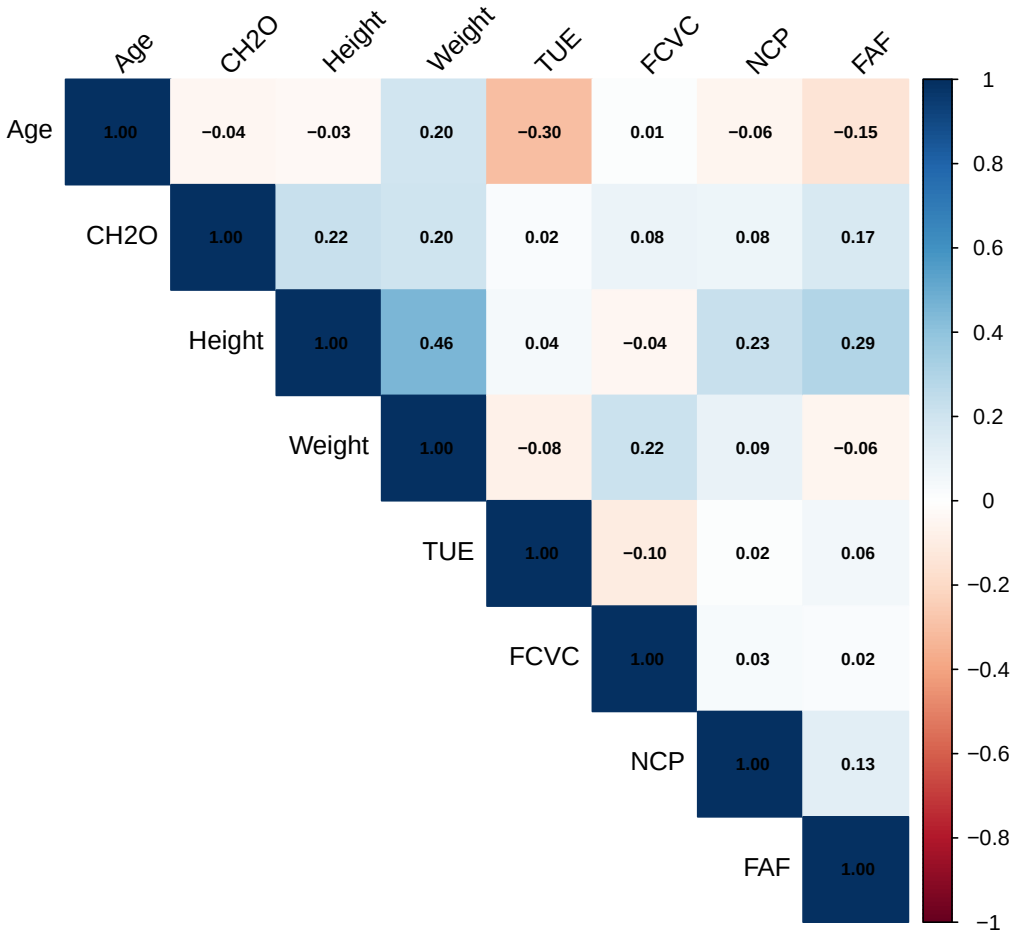

Density Plot of Weight
Density
0.000
0.005
0.010
0.015
50
100
150
Weight (kg)


Weight vs Age
Weight (kg)
40
80
120
160
20
30
40
50
60
Age


Weight vs Height
Weight (kg)
40
80
120
160
1.5
1.6
1.7
1.8
1.9
2.0
Height (m)


Weight by Gender
Weight (kg)
40
80
120
160
Female
Male
Gender


Weight by Family History
Weight (kg)
40
80
120
160
no
yes
Family History



Our dataset consists primarily of young adults, with age centered in the early 20s and a right-skewed tail extending into older ages. Height is tightly distributed around approximately 1.70 meters, while weight shows substantial variability, ranging from under 40 kg to over 170 kg. Weight has a clear linear relationship with age and height. The boxplots showed that individuals with a family history of overweight display noticeably higher weights compared to those without such a history. The correlation matrix showed that height had the strongest correlation with weight (positive relationship). Age shows a moderate positive correlation with weight. FAF (physical activity) shows negative correlation with Weight as expected. Most predictor correlations are weak, suggesting minimal multicollinearity issues.

2.3 Initial Full Linear Model

```
##
## Call:
## lm(formula = Weight ~ . - NObeyesdad, data = obesity)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -54.402 -12.009   1.453  10.423  78.626
##
## Coefficients:
##                                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  -193.98452   19.84114  -9.777  < 2e-16 ***
## GenderMale                     -4.50309    1.03911  -4.334 1.54e-05 ***
```

```
## Age                                  0.96844    0.08089  11.972  < 2e-16 ***
## Height                             123.17361    5.95264  20.692  < 2e-16 ***
## family_history_with_overweightyes   18.10795    1.11988  16.170  < 2e-16 ***
## FAVCyes                              4.58561    1.27390   3.600 0.000326 ***
## FCVC                                 8.79018    0.75133  11.699  < 2e-16 ***
## NCP                                  0.52983    0.51261   1.034 0.301452
## CAECFrequently                      -9.31993    2.61909  -3.558 0.000381 ***
## CAECno                               4.77954    3.72217   1.284 0.199261
## CAECSometimes                        9.87700    2.42134   4.079 4.69e-05 ***
## SMOKEyes                            -0.67575    2.65232  -0.255 0.798922
## CH2O                                 1.62481    0.65578   2.478 0.013303 *
## SCCyes                              -4.90176    1.87924  -2.608 0.009163 **
## FAF                                 -2.52326    0.48681  -5.183 2.39e-07 ***
## TUE                                 -1.33177    0.66448  -2.004 0.045175 *
## CALCFrequently                     -11.92271   17.39806  -0.685 0.493238
## CALCno                             -14.47612   17.29010  -0.837 0.402550
## CALCSometimes                       -7.33806   17.30367  -0.424 0.671555
## MTRANSBike                           5.09961    6.55557   0.778 0.436714
## MTRANSMotorbike                     12.33709    5.27086   2.341 0.019346 *
## MTRANSPublic_Transportation         14.22821    1.17708  12.088  < 2e-16 ***
## MTRANSWalking                        4.20865    2.60691   1.614 0.106590
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17.03 on 2064 degrees of freedom
## Multiple R-squared:  0.5816, Adjusted R-squared:  0.5771
## F-statistic: 130.4 on 22 and 2064 DF,  p-value: < 2.2e-16
```

Model Interpretation

The initial full model includes all available predictors except the obesity category variable (NObeyesdad), which is directly derived from weight and BMI, making it inappropriate as a predictor. Weight (kg) is the response variable.

Coefficient Interpretation:

Quantitative Predictors:

- **Height**: Height has the strongest association with weight. A one-meter increase in height is associated with an average increase of approximately 123 kg in weight.
- **Age**: Age is positively associated with weight. A one-year increase in age corresponds to an average increase of approximately 0.97 kg in weight.
- **Physical Activity Frequency (FAF)**: Significantly negatively associated with weight. Each one-unit increase in physical activity frequency corresponds to an average decrease of 2.52 kg in weight.
- **Daily Water Consumption (CH2O)**: Shows a small but statistically significant positive association (1.62 kg per unit), suggesting that water intake alone does not strongly differentiate body weight once other behaviors are accounted for.

Categorical Predictors:

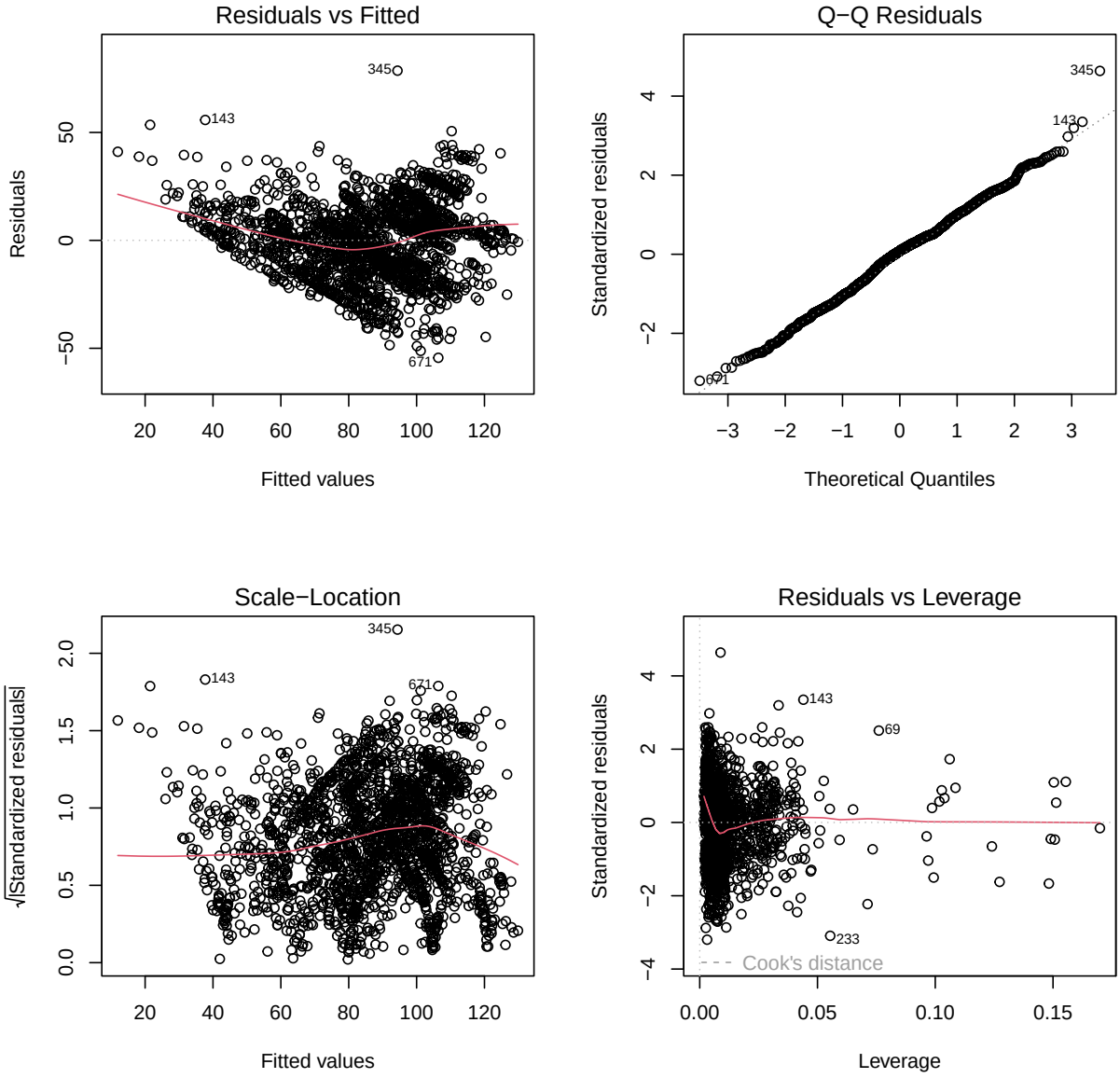
- **Gender (Male)**: Males weigh on average 4.5 kg less than females when all other factors are held constant, contrary to typical population patterns, possibly due to other confounding factors in this dataset.

- **Family History of Overweight**: Individuals with a family history of overweight weigh approximately 18.11 kg more on average than those without such history, highlighting the strong influence of genetic and environmental factors.
- **Frequent Consumption of High-Calorie Foods (FAVC)**: Associated with a 4.59 kg increase in weight compared to those who don't frequently consume high-calorie foods.
- **Transportation Mode**:
  - Public transportation users weigh approximately 14.23 kg more than automobile users
  - Motorbike users weigh about 12.34 kg more than automobile users
  - Walking shows a non-significant increase of 4.21 kg compared to automobile use

R-squared Interpretation:

The R-squared value of 0.5816 indicates that approximately 58.16% of the variation in weight can be explained by the predictors in our model. The adjusted R-squared is 0.5771, which we will use as a baseline for model comparison. The small difference between R-squared and adjusted R-squared suggests that, despite the large number of predictors, the model is not overly inflated by unnecessary variables.

Model Diagnostics




```
## [1] "Variance Inflation Factors:"

##  [1] 4.000000 3.000000 3.000000 2.212661 1.941953 1.908563 1.893853 1.487502
##  [9] 1.393540 1.381507

##
## Shapiro-Wilk test p-value: 4.871521e-05

## Breusch-Pagan test p-value: 1.175803e-53
```

Cross-Validation

Model	CV_RMSE	CV_R_squared	CV_MAE
Full Model	17.11762	0.5732762	13.60475

Based on the diagnostics, the full model satisfies most of the regression assumptions. The Fitted vs Residuals plot shows relatively random scatter around zero. Linearity is reasonably satisfied. The Q-Q plot shows mild departures from normality in the tails, and the Shapiro-Wilk test rejects normality. Multicollinearity is not a concern (all VIFs < 5). The Breusch–Pagan test indicates heteroscedasticity. Given the reasonable diagnostics and baseline predictive performance, the next section explores alternative model specifications aimed at addressing the observed heteroscedasticity.

2.4 Model Improvement Methods

Variable Selection Using Stepwise Methods

Model	CV_RMSE	CV_R_squared	CV_MAE
Backward AIC	17.1283	0.5741715	13.63747

Model	CV_RMSE	CV_R_squared	CV_MAE
Forward AIC	17.09096	0.5747511	13.60911

Model	CV_RMSE	CV_R_squared	CV_MAE
Both-direction AIC	17.12083	0.574863	13.63512

Model	CV_RMSE	CV_R_squared	CV_MAE
Backward BIC	17.13498	0.5723123	13.67278

```
## Model Comparison - Adjusted R-squared:
```

```
## Full Model: 0.5771166
```

```
## Backward AIC: 0.5772936
```

```
## Forward AIC: 0.5772936
```

```
## Backward BIC: 0.5744975
```

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
2069	603895.4	NA	NA	NA	NA
2066	599057.2	3	4838.2395	5.5596364	0.0008476

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
2064	598727.8	2	329.3904	0.5677553	0.5668848

Variable selection using stepwise methods showed minimal improvement in predictive performance. AIC-based stepwise selection (backward, forward, and both directions) retained most predictors from the full model. BIC-based selection removed several predictors. Removing predictors did not meaningfully improve model fit. The variable selection models all had RMSE around 17.1 kg.

Polynomial Transformations

```
## [1] 0.5933315
```

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
2066	599057.2	NA	NA	NA	NA
2063	597512.1	3	1545.053	1.778175	0.149236

Model	CV_RMSE	CV_R_squared	CV_MAE
Polynomial	16.78698	0.5902946	13.18181

Adding quadratic terms for Height, Age, and Physical Activity (FAF) increased adjusted R squared to approximately 0.593, representing a noticeable improvement over the initial linear model.

Interaction Terms

```
## Adjusted R-squared with interactions:

## Height * Gender: 0.5913179

## Age * Family History: 0.5775538

## Height * Age: 0.5844035
```

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
2066	599057.2	NA	NA	NA	NA
2063	578341.0	3	20716.18	24.63222	0

Model	CV_RMSE	CV_R_squared	CV_MAE
Height * Gender	16.84767	0.5886097	13.62921

The Height × Gender interaction improved the model fit from the baseline model and the stepwise models. Allowing the relationship between height and weight to differ by gender meaningfully improves model performance.

Combined Model: AIC Selection + Interaction

```
## Combined Model Formula:
```

```
## Weight ~ Height * Gender + Age + family_history_with_overweight +
##     FAVC + FCVC + CAEC + CH2O + SCC + FAF + TUE + CALC + MTRANS
```

```
## Adjusted R-squared: 0.5916482
```

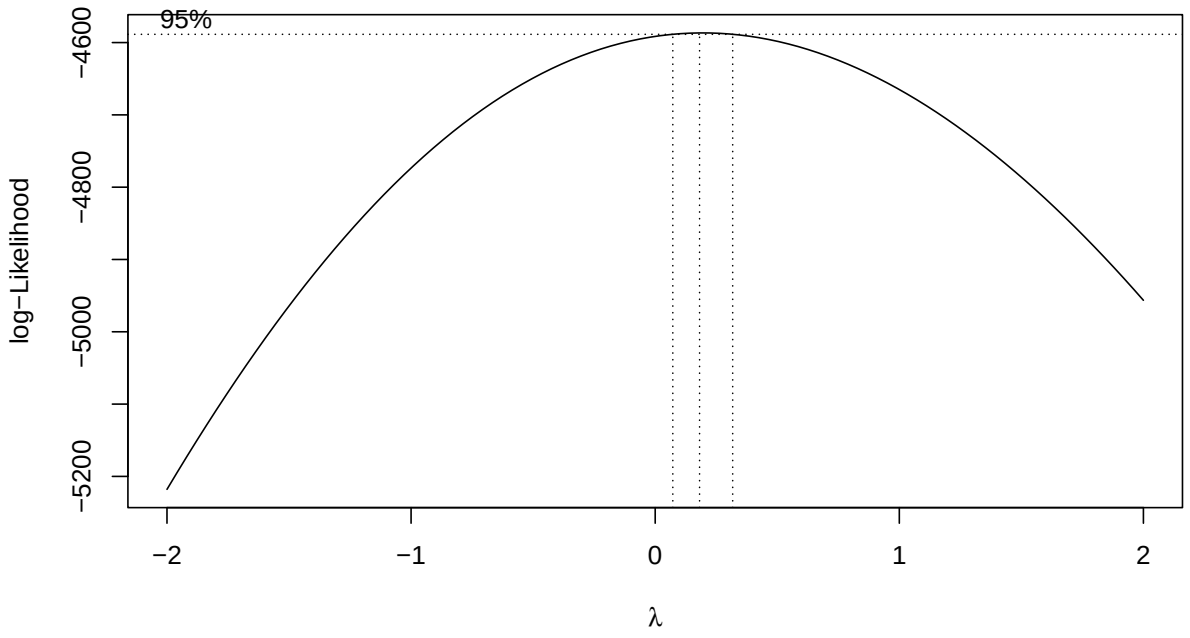
```
## Residual Standard Error: 16.73658 kg
```

```
## F-statistic: 144.9209
```

```
## Degrees of Freedom: 22 used, 2065 residual
```

The combined model retains important predictors from AIC selection and includes the best interaction. The combined approach achieves an adjusted R squared of 0.592 and a cross validated RMSE of 16.79 kg.

Box-Cox Transformation




```
## Optimal lambda for Box-Cox transformation: 0.1818182
```

```
## Adjusted R-squared with Box-Cox transformation: 0.5967893
```

The optimal Box-Cox is 0.18. The transformed model's adjusted R^2 (0.597) is on the transformed scale and isn't directly comparable to the other models, and it makes coefficients harder to interpret in kg, so we kept Weight untransformed.

intercept	RMSE	Rsquared	MAE	RMSESD	RsquaredSD	MAESD
TRUE	17.11762	0.5732762	13.60475	0.8116383	0.0364084	0.6415173

Model Comparison

```
##                       Model     RMSE  Rsquared      MAE
## 1 Combined_AIC_Interaction 16.78573 0.5897206 13.56997
## 2               Polynomial 16.80014 0.5910337 13.18147
## 3   Height_Gender_Interact 16.83179 0.5881586 13.60578
## 4             Backward_AIC 17.07558 0.5766074 13.61372
## 5               Full Model 17.11762 0.5732762 13.60475
## 6             Backward_BIC 17.15863 0.5725245 13.69526
```

The Combined model shows best balance across all metrics. The interaction and polynomial models both performed well. Backward BIC performed the worst.

3. CONCLUSION

3.1 Final Model Selection

Based on comprehensive model comparison using adjusted R-squared, cross-validation RMSE, and mean absolute error, we select the model that combines AIC backward selection with Height-Gender interaction as our final model.

```
##
## Call:
## lm(formula = Weight ~ Height * Gender + Age + family_history_with_overweight +
##     FAVC + FCVC + CAEC + CH2O + SCC + FAF + TUE + CALC + MTRANS,
##     data = obesity)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -55.824 -12.084   2.473  11.195  84.242
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      -270.32741   21.35564 -12.658  < 2e-16 ***
## Height                            170.04920    7.86181  21.630  < 2e-16 ***
## GenderMale                        149.53909   17.98941   8.313  < 2e-16 ***
## Age                                 0.99507    0.07883  12.624  < 2e-16 ***
## family_history_with_overweightyes  17.74610    1.10087  16.120  < 2e-16 ***
## FAVCyes                             3.86847    1.25123   3.092 0.002017 **
## FCVC                                8.01899    0.74370  10.783  < 2e-16 ***
## CAECFrequently                     -9.34332    2.57326  -3.631 0.000289 ***
## CAECno                              1.27367    3.67193   0.347 0.728728
## CAECSometimes                       9.39283    2.37728   3.951 8.04e-05 ***
```

```
## CH2O                                1.59104    0.64325   2.473 0.013463 *
## SCCyes                             -2.88916    1.85966  -1.554 0.120435
## FAF                                -2.76140    0.47851  -5.771 9.08e-09 ***
## TUE                                -1.78268    0.65403  -2.726 0.006471 **
## CALCFrequently                     -8.46250   17.07705  -0.496 0.620266
## CALCno                            -10.64099   16.97165  -0.627 0.530738
## CALCSometimes                      -2.77472   16.98145  -0.163 0.870222
## MTRANSBike                          3.57402    6.44438   0.555 0.579232
## MTRANSMotorbike                    10.87355    5.17722   2.100 0.035826 *
## MTRANSPublic_Transportation        13.57726    1.15109  11.795  < 2e-16 ***
## MTRANSWalking                       4.61102    2.55960   1.801 0.071776 .
## Height:GenderMale                 -90.56904   10.55521  -8.581  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 16.74 on 2065 degrees of freedom
## Multiple R-squared:  0.5958, Adjusted R-squared:  0.5916
## F-statistic: 144.9 on 21 and 2065 DF,  p-value: < 2.2e-16
```

3.2 Final Model Analysis

Model Performance Metrics

```
## Final Model Performance:

## R-squared: 0.5957591

## Adjusted R-squared: 0.5916482

## Residual Standard Error: 16.73658

## AIC: 17707.18

## BIC: 17836.98

##
## Comparison with Initial Full Model:

## Initial Adjusted R-squared: 0.5771166

## Improvement: 0.01453155

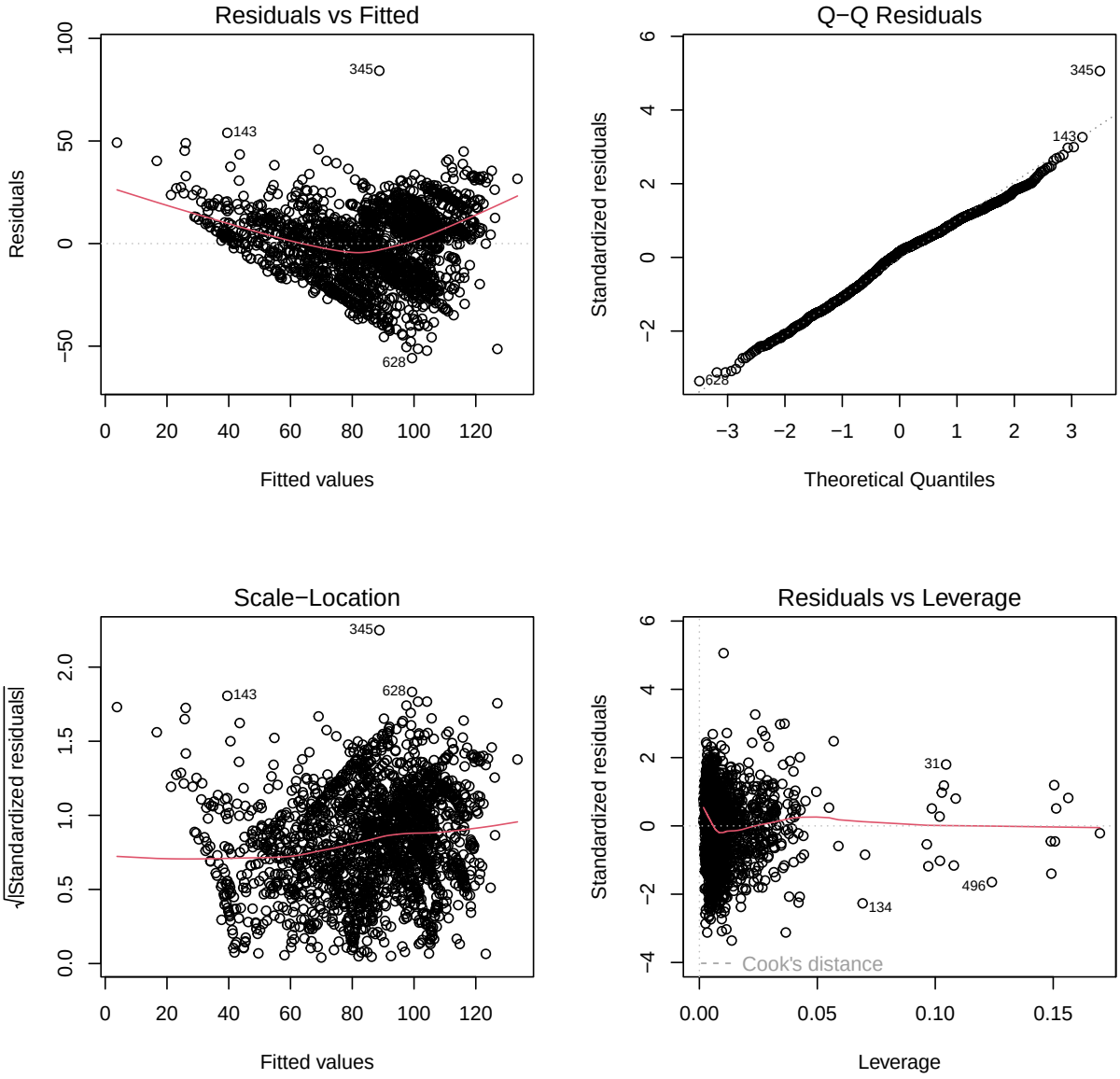
## Initial AIC: 17781.15

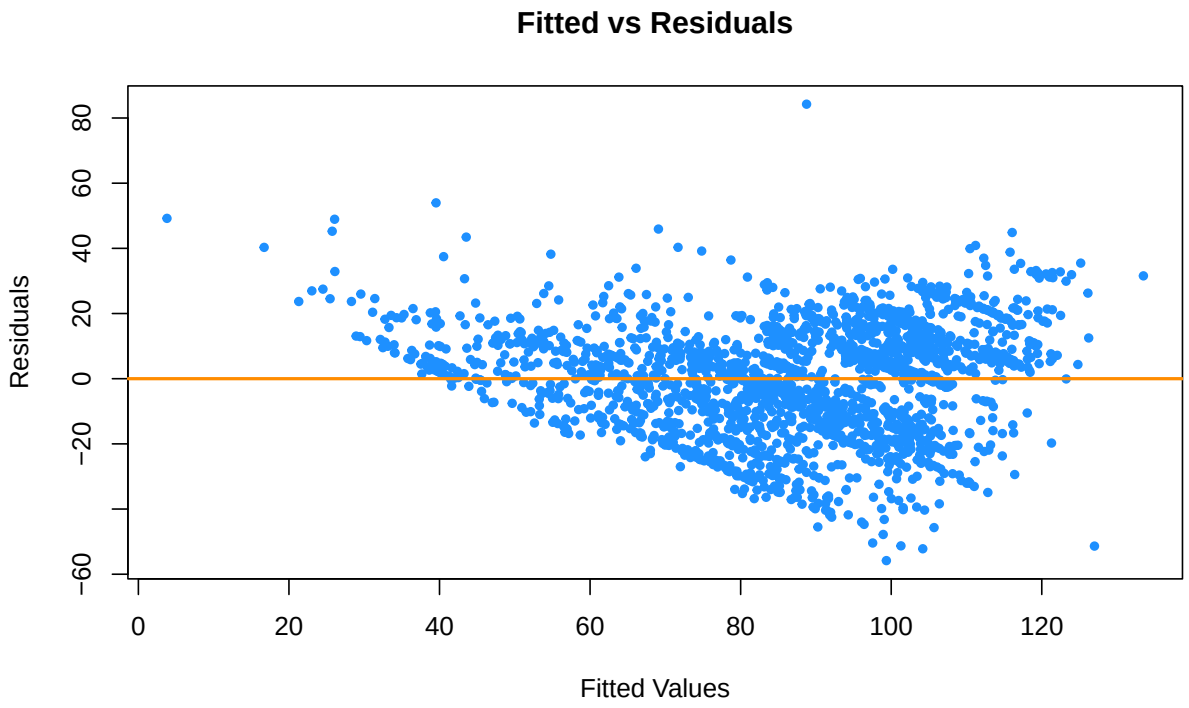
## Initial BIC: 17916.59
```

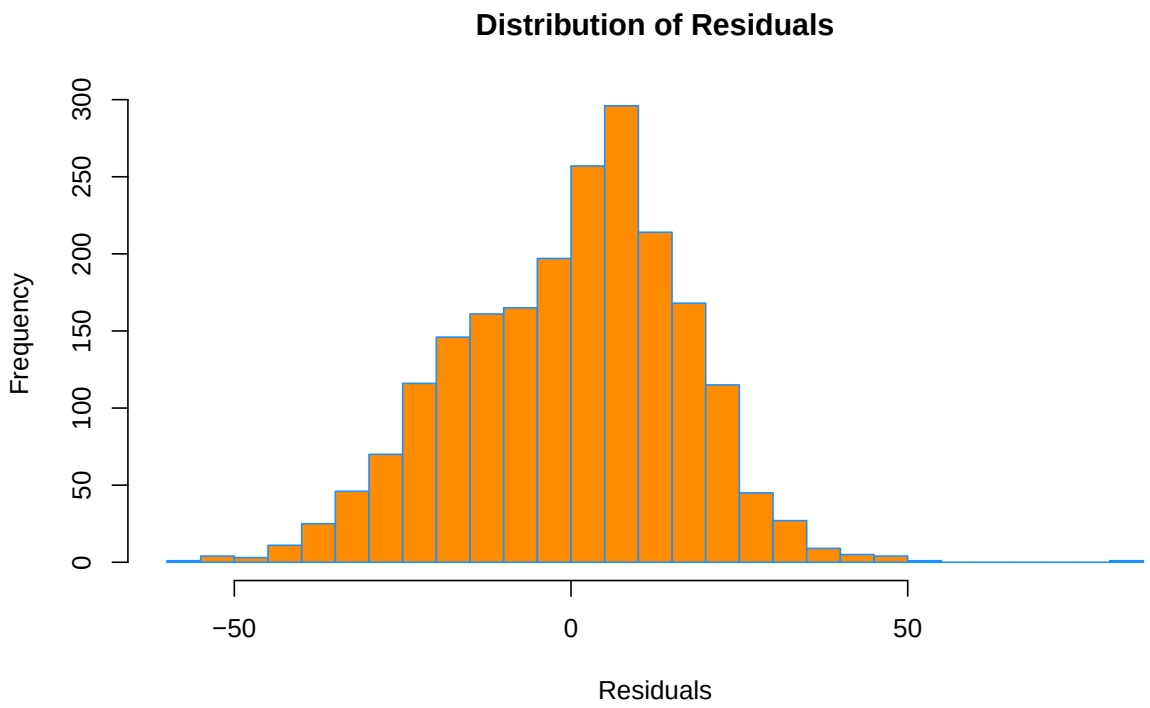
Selection Rationale:

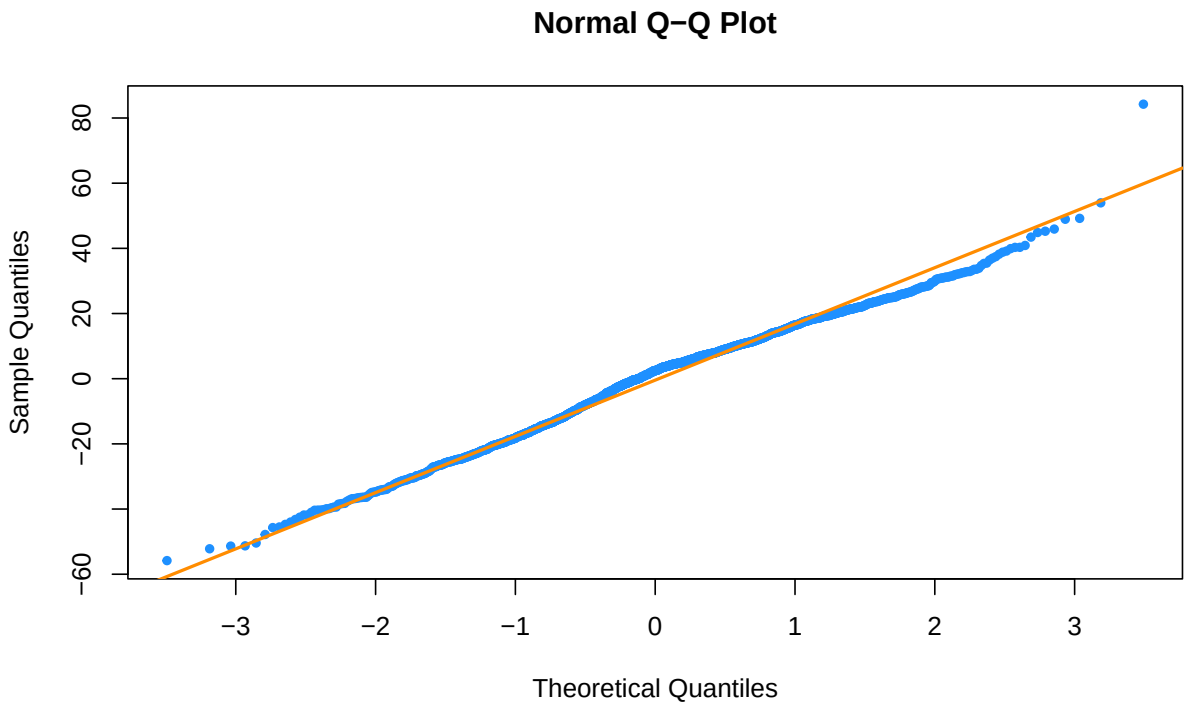
The Combined Model was chosen over alternatives because it produced the lowest cross-validated RMSE (16.8 kg), improving slightly upon both the full model and AIC/BIC models. Incorporating the Height $\times$ Gender interaction yielded modest gains in predictive performance.

Final Model Diagnostics




Fitted vs Residuals
Residuals
80
60
40
20
0
−20
−60
0
20
40
60
80
100
120
Fitted Values


Distribution of Residuals
Frequency
300
250
200
150
100
50
0
−50
0
50
Residuals




```
## Final Model Assumption Tests:

## Shapiro-Wilk test p-value: 7.366043e-11

## Breusch-Pagan test p-value: 1.703283e-49

##
## VIF values (top 5):

## [1] 645.30747 602.74511  25.40290  24.55087   4.00000
```

Influential Observations

```
## Number of potentially influential observations: 85

##
## Model without top influential points:
## Original Adjusted R-squared: 0.5916482
## Clean Adjusted R-squared: 0.5942934
```

Cross-Validation of Final Model

```
## 10-Fold Cross-Validation for Final Model:

## RMSE: 16.81441
```

```
## R-squared: 0.5879523

## MAE: 13.59807

##
##
## Comparison with Baseline:

## Baseline RMSE: 17.0961

## Final RMSE: 16.81441

## Improvement: 0.2816959
```

Linearity is reasonably satisfied. Residuals show some departure from normality in the tails and clear heteroscedasticity (Breusch–Pagan $p < 0.001$). About 4% of points are flagged as influential, but removing the top ones barely changes adjusted R^2.

Model Interpretation

```
## Coefficient Estimates:

##                                     Estimate  Std. Error     t value
## (Intercept)                      -270.3274062 21.35564094 -12.6583607
## Height                            170.0491960  7.86181222  21.6297708
## GenderMale                        149.5390938 17.98940636   8.3126197
## Age                                 0.9950709  0.07882673  12.6235207
## family_history_with_overweightyes  17.7460962  1.10086575  16.1201274
## FAVCyes                             3.8684726  1.25123393   3.0917261
## FCVC                                8.0189875  0.74370118  10.7825397
## CAECFrequently                     -9.3433198  2.57326030  -3.6309268
## CAECno                              1.2736650  3.67193448   0.3468649
## CAECSometimes                       9.3928316  2.37727599   3.9510901
## CH2O                                1.5910361  0.64325322   2.4734211
## SCCyes                             -2.8891581  1.85966342  -1.5535919
## FAF                                -2.7613965  0.47850731  -5.7708555
## TUE                                -1.7826814  0.65403124  -2.7256824
## CALCFrequently                     -8.4624997 17.07704537  -0.4955482
## CALCno                            -10.6409883 16.97165325  -0.6269860
## CALCSometimes                      -2.7747184 16.98145180  -0.1633970
## MTRANSBike                          3.5740158  6.44437689   0.5545945
## MTRANSMotorbike                    10.8735520  5.17721521   2.1002704
## MTRANSPublic_Transportation        13.5772551  1.15108984  11.7951307
## MTRANSWalking                       4.6110194  2.55959565   1.8014640
## Height:GenderMale                 -90.5690389 10.55520566  -8.5805092
##                                     Pr(>|t|)
## (Intercept)                      2.007302e-35
## Height                           1.093883e-93
## GenderMale                       1.675006e-16
## Age                              3.028581e-35
## family_history_with_overweightyes 3.708066e-55
```

```
## FAVCyes                          2.016553e-03
## FCVC                             2.071505e-26
## CAECFrequently                   2.892826e-04
## CAECno                           7.287282e-01
## CAECSometimes                    8.040961e-05
## CH2O                             1.346255e-02
## SCCyes                           1.204350e-01
## FAF                              9.078991e-09
## TUE                              6.471046e-03
## CALCFrequently                   6.202659e-01
## CALCno                           5.307377e-01
## CALCSometimes                    8.702218e-01
## MTRANSBike                       5.792321e-01
## MTRANSMotorbike                  3.582607e-02
## MTRANSPublic_Transportation      4.025470e-31
## MTRANSWalking                    7.177559e-02
## Height:GenderMale                1.824694e-17
```

- **Age:** For each additional year of age, weight increases by 0.995 kg on average, holding all else constant.
- **Height:** The effect of height on weight depends on gender.
  - **For Females:** Each additional meter increases weight by 170.05 kg on average.
  - **For Males:** Each additional meter increases weight by 79.48 kg on average.
- **Family history:** Having family history of overweight changes weight by 17.75 kg on average.

3.3 Concluding Thoughts

Model Interpretation

Our final multiple linear regression model explains approximately 59.2% of the variance in body weight. The model performs well with a 10-fold cross-validated RMSE of 16.81 kg, indicating reasonable predictive accuracy.

The highly significant F-statistic confirms that our model is significantly better than an intercept-only model. This suggests that the selected lifestyle and demographic factors collectively have meaningful relationships with body weight.

Key Findings

1. **Height-Gender Interaction**: The significant interaction between height and gender indicates that the relationship between height and weight differs between males and females. Females show a steeper weight increase per unit height compared to males (about 1.7 kg per cm for females vs. 0.79 kg per cm for males).

2. **Physical Characteristics**: Both height and age are significant predictors of weight. Taller individuals and older individuals tend to weigh more.

3. **Lifestyle Factors**: Physical activity frequency (FAF) is negatively associated with weight (about -2.76 kg per unit), while vegetable consumption (FCVC) is positively associated (about $+8.02$ kg per unit).

4. **Family History**: Family history with overweight is a significant predictor, suggesting genetic or shared environmental factors.

5. **Model Performance**: With an adjusted R-squared of 0.5916, our model explains approximately 59.2% of the variation in weight, which is quite good for a biological/behavioral model.

Limitations

1. **Sample Representativeness**: Data is limited to three Latin American countries, potentially limiting generalizability to other populations.
2. **Residual Patterns**: Even with the interaction term, some heteroscedasticity remains, suggesting the relationship between predictors and weight may be more complex than captured by linear models.
3. **Multicollinearity**: VIF values are generally acceptable (< 5), indicating multicollinearity is not a severe concern. The interaction term creates expected correlation with component variables.
4. **Model Complexity**: With n = 2087 observations and 21 predictors (including interaction), we have adequate sample size. The model may not capture all non-linear relationships.
5. **Influential Observations**: About 4.1% of observations are influential. These points may affect model fit but are not excessive.

Recommendations

1. The model performs well enough for general weight prediction and could be useful for understanding which factors relate to weight.
2. Future improvements could include:
    - Non-linear modeling approaches (Neural Networks, random forests)
    - Additional predictors such as genetic markers or detailed dietary information
    - Geographic and cultural factors beyond the current dataset

4. APPENDIX

4.1 Important Models

```r
# Fit the full model with all predictors except NObeyesdad
# because it directly relates to weight
full_model <- lm(Weight ~ . - NObeyesdad, data = obesity)
summary(full_model)

# Backward selection using AIC
model_backward_aic <- step(full_model, direction = "backward", trace = 0)
#cat("Backward Selection (AIC) - Variables retained:\n")
#names(coef(model_backward_aic))
get_cv_metrics(formula(model_backward_aic), obesity, "Backward AIC")

# Forward selection
null_model <- lm(Weight ~ 1, data = obesity)
model_forward_aic <- step(null_model,
scope = list(lower = null_model, upper = full_model), direction = "forward", trace = 0)
#cat("\n\nForward Selection (AIC) - Variables selected:\n")
```

```r
#names(coef(model_forward_aic))
get_cv_metrics(formula(model_forward_aic), obesity, "Forward AIC")

# Both directions
model_both_aic <- step(full_model, direction = "both", trace = 0)
get_cv_metrics(formula(model_both_aic), obesity, "Both-direction AIC")

# BIC
n <- nrow(obesity)
model_backward_bic <- step(full_model, direction = "backward", k = log(n), trace = 0)

#cat("\n\nBackward Selection (BIC) - Variables retained:\n")
#names(coef(model_backward_bic))
get_cv_metrics(formula(model_backward_bic), obesity, "Backward BIC")

# Add polynomial terms for continuous predictors with strong relationships
model_poly <- lm(Weight ~ poly(Height, 2) + poly(Age, 2) + poly(FAF, 2) + Gender +
family_history_with_overweight + FAVC + FCVC + NCP + CAEC + SMOKE + CH2O +
SCC + TUE + CALC + MTRANS, data = obesity)

summary(model_poly)$adj.r.squared

# Test if polynomial terms improve the model
model_height_poly <- lm(Weight ~ Height + I(Height^2) + Age + Gender +
family_history_with_overweight + FAVC + FCVC + NCP +
CAEC + SMOKE + CH2O + SCC + FAF + TUE + CALC +
MTRANS, data = obesity)

anova(model_backward_aic, model_height_poly)

get_cv_metrics(formula(model_poly), obesity, "Polynomial")

# Combine AIC backward selection with best interaction
model_combined <- lm(Weight ~ Height * Gender + Age +
                     family_history_with_overweight + FAVC + FCVC +
                     CAEC + CH2O + SCC + FAF + TUE + CALC + MTRANS,
                     data = obesity)

cat("Combined Model Formula:\n")
print(formula(model_combined))
#summary(model_combined)
summary_combined <- summary(model_combined)

cat("Adjusted R-squared:", summary_combined$adj.r.squared, "\n")
cat("Residual Standard Error:", summary_combined$sigma, "kg\n")
cat("F-statistic:", summary_combined$fstatistic[1], "\n")
cat("Degrees of Freedom:", summary_combined$df[1], "used,",
    summary_combined$df[2], "residual\n")
```

4.2 Selected Final Model

```r
# Final selected model
final_model <- lm(Weight ~ Height * Gender + Age +
                    family_history_with_overweight + FAVC + FCVC +
                    CAEC + CH2O + SCC + FAF + TUE + CALC + MTRANS,
                    data = obesity)

summary(final_model)
```

4.3 Alternative Final Model

```
## Alternative Model (No Interaction):

##
## Call:
## lm(formula = Weight ~ Age + Height + Gender + family_history_with_overweight +
##     FAVC + FCVC + CH2O + FAF, data = obesity)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -57.712 -12.430   1.914  12.757  64.044
##
## Coefficients:
##                                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     -197.56295   10.06581 -19.627  < 2e-16 ***
## Age                                0.52270    0.06804   7.683 2.38e-14 ***
## Height                           131.36831    6.33039  20.752  < 2e-16 ***
## GenderMale                        -5.60417    1.13870  -4.922 9.26e-07 ***
## family_history_with_overweightyes  22.11862    1.19152  18.563  < 2e-16 ***
## FAVCyes                            8.91817    1.36450   6.536 7.93e-11 ***
## FCVC                               9.51128    0.82722  11.498  < 2e-16 ***
## CH2O                               2.97613    0.71814   4.144 3.55e-05 ***
## FAF                               -4.23354    0.52948  -7.996 2.12e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19.04 on 2078 degrees of freedom
## Multiple R-squared:  0.4737, Adjusted R-squared:  0.4717
## F-statistic: 233.8 on 8 and 2078 DF,  p-value: < 2.2e-16

##
##
## Comparison with Final Model:

## Final Model (with interaction):

##   AIC: 17707.18

##   Adj R²: 0.5916482
```

```
## Alternative Model (no interaction):

##   AIC: 18231.95

##   Adj R²: 0.4716514

## Nested F-test for interaction term:

## Analysis of Variance Table
##
## Model 1: Weight ~ Age + Height + Gender + family_history_with_overweight +
##     FAVC + FCVC + CH2O + FAF
## Model 2: Weight ~ Height * Gender + Age + family_history_with_overweight +
##     FAVC + FCVC + CAEC + CH2O + SCC + FAF + TUE + CALC + MTRANS
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1   2078 753122
## 2   2065 578434 13    174688 47.972 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Observations:

- Alternative model without interaction has higher AIC and lower Adjusted R^2
- Final model with interaction provides statistically better fit
- Nested F-test confirms interaction term is significant
- Interaction meaningfully improves model and should be retained

---

4.4 Coefficient Confidence Intervals

```r
# 95% Confidence intervals for final model coefficients
ci_final <- confint(final_model)
significant_coefs <- summary(final_model)$coefficients[, "Pr(>|t|)"] < 0.05

cat("Significant Coefficients (p < 0.05) with 95% CI:\n\n")
for(i in which(significant_coefs)) {
  coef_name <- names(coef(final_model))[i]
  cat(sprintf("%-30s: %7.4f [%7.4f, %7.4f]\n",
              coef_name,
              coef(final_model)[i],
              ci_final[i, 1],
              ci_final[i, 2]))
}
```

End of Report